Vehicle Architecture-Driven Design and Optimization of an 800V Three-Level Flying Capacitor Traction Inverter

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***Abstract*—This paper presents a comprehensive design, simulation, and switching loss optimization of a Three-Level Flying Capacitor (3L-FC) inverter for an 800V Electric Vehicle (EV) traction application. The selection of the FC topology over the Neutral Point Clamped (NPC) alternative is detailed, highlighting advantages in thermal symmetry, component count, and natural balancing under dynamic EV loads. Component selection criteria for the 1200V SiC modules and high-ripple film capacitors are established based on industry standards. By mapping vehicle speed directly to inverter electrical parameters through mechanical drivetrain characteristics, a full driving spectrum analysis was conducted. Utilizing the Graovac-Purschel analytical method and datasheet parameters from a commercial 1200V, 450A SiC module, the switching losses were quantified. Results show that an adaptive switching frequency schedule (5 kHz for urban, 20 kHz for highway) yields a 75% reduction in switching losses during high-current urban driving compared to a fixed 20 kHz baseline. Time-domain simulations validate the design, demonstrating successful natural capacitor balancing, a low Total Harmonic Distortion (THD) of 1.92% at urban speeds, proper pre-charge transient handling, and safe thermal operation.**

I. INTRODUCTION

The rapid evolution of Electric Vehicles (EVs) is driving a shift from traditional 400V to 800V DC bus architectures. This transition enables faster charging times, reduced cable weight, and higher overall system efficiency [1]. However, the increased DC link voltage presents significant challenges for the traction inverter, primarily concerning semiconductor voltage ratings, electromagnetic interference (EMI), and the dv/dt stress imposed on motor windings [2].

Traditional two-level voltage source inverters (2L-VSI) require semiconductor devices with blocking voltages of at least 1200V. While Silicon Carbide (SiC) MOSFETs meet this requirement, their fast switching speeds exacerbate dv/dt stress, leading to premature motor insulation failure [3]. Multilevel inverters offer a compelling solution by synthesizing the output voltage from multiple discrete levels, inherently reducing the voltage step size and improving the harmonic profile.

II. TOPOLOGY SELECTION: FC VS. NPC

When designing a three-level traction inverter for an 800V EV architecture, the two primary candidates are the Neutral Point Clamped (NPC) and the Flying Capacitor (FC) topologies. A rigorous comparative analysis dictates the selection of the FC topology for this application based on several critical factors unique to EV traction [4].

A. Thermal Symmetry and Component Count

A major disadvantage of the NPC topology is the unequal distribution of conduction and switching losses among its semiconductor devices. The outer switches in an NPC inverter process different current profiles than the inner switches, leading to asymmetric thermal stress. This complicates cooling system design and degrades long-term reliability [5]. In contrast, the FC topology distributes voltage and current stresses symmetrically across all four switches in a phase leg. Every switch blocks exactly half the DC bus voltage (Vdc/2), allowing the use of identical power modules throughout the design. Furthermore, the FC topology eliminates the need for the clamping diodes required in the NPC, removing components from the conduction path and thereby reducing overall conduction losses [6].

B. Dynamic Load Balancing

EV traction drives are characterized by highly dynamic, rapidly fluctuating load profiles. The NPC topology relies on the neutral point of the DC-link capacitor bank, which is notoriously difficult to keep balanced under transient conditions or varying power factors without injecting complex zero-sequence voltage commands [5]. The FC topology, however, utilizes redundant switching states to naturally balance the flying capacitor voltage. By actively selecting between redundant states based on the phase current direction, the flying capacitor remains balanced at Vdc/2 regardless of the load power factor, making it highly robust for EV drive cycles.

C. Trade-offs and Mitigations

The primary drawback of the FC topology is the requirement for bulky, high-voltage capacitors in each phase leg, which add volume and cost compared to the NPC [4]. Additionally, a dedicated pre-charge circuit is required at startup to prevent catastrophic overvoltage across the inner switches. However, in modern EV architectures, pre-charging contactors are already mandated for the main DC-link, and advancements in automotive-grade film capacitors have significantly reduced the volume penalty, making the FC topology highly viable.

III. COMPONENT SELECTION

The physical components were selected based on rigorous automotive industry standards to ensure reliability under the 800V architecture.

A. Semiconductor Switches

The Wolfspeed CAB450M12XM3 SiC half-bridge power module was selected for the active switches. The 1200V blocking rating provides a mandatory 50% safety margin over the 800V nominal bus (400V per switch in the 3L-FC configuration) to withstand transient spikes. The 450A continuous current rating provides a robust 3.3x margin over the worst-case peak urban phase current (136A), accounting for transient overloads, high-temperature derating, and safe operating area (SOA) constraints [7]. SiC technology was chosen over Silicon IGBTs due to its order-of-magnitude lower switching energy and absence of tail current, enabling the high switching frequencies (up to 20 kHz) required for high-speed motor operation without prohibitive thermal losses.

B. Flying Capacitors

The flying capacitor (Cfc) must be sized to maintain voltage ripple within a 5% budget (20V on a 400V reference) under the worst-case operating condition, which occurs at maximum current and minimum switching frequency. Using the sizing equation C = Ipk / (2 * fsw_min * ΔVfc), with Ipk = 136A, fsw_min = 5 kHz, and ΔVfc = 20V, the required capacitance is 680 µF. Automotive-grade metallized polypropylene film capacitors (e.g., TDK EPCOS or KEMET C4AQ series) are specified due to their exceptionally low Equivalent Series Resistance (ESR < 5 mΩ), high ripple current capability (>100A RMS), and self-healing properties, which are critical for the reliability of the FC topology [8].

IV. VEHICLE ARCHITECTURE AND OPERATING POINTS

To accurately model inverter losses and performance, the electrical operating points must be mapped from the vehicle's mechanical state. The test vehicle architecture is defined by an 800V DC bus, a single-speed reduction gear (ratio G = 8.19), a tyre radius (r) of 0.315 m, and an 8-pole Permanent Magnet Synchronous Motor (pole pairs p = 4).

A. Speed to Frequency Mapping

The fundamental electrical frequency (fe) required by the motor is directly proportional to the vehicle speed (v, in m/s):

$$fe = (v / r) * (G / 2\pi) * p$$

Urban driving (40 km/h) requires a relatively low electrical frequency of 184 Hz, while highway cruising (120 km/h) demands 552 Hz.

B. Current Profile Constraints

The inverter was modeled driving an RL load (R=1.3Ω, L=2mH). Because the load impedance increases with frequency, the peak phase current (Ipk) is inversely proportional to vehicle speed. Urban driving represents the highest current stress (136 A at 40 km/h), while highway driving draws significantly less current (51 A at 120 km/h).

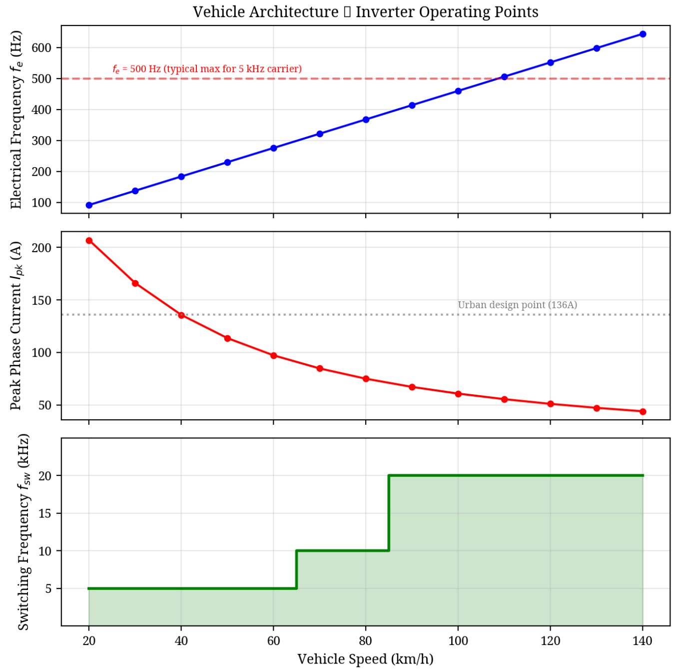

Fig. 1. Mapping of vehicle speed to inverter electrical frequency and peak phase current, showing the adaptive switching frequency schedule.

V. ANALYTICAL SWITCHING LOSS OPTIMIZATION

To quantify the efficiency impact of the drive frequency, the industry-standard analytical method proposed by Graovac and Purschel [9] was utilized. This method calculates the duty-cycle-weighted average switching loss over a fundamental cycle.

A. The Graovac-Purschel Method for 3L-FC

The total switching energy (Esw = Eon + Eoff) of a SiC MOSFET scales approximately linearly with commutated current and voltage. By integrating the switching energy over a sinusoidal current half-wave, the average switching power loss per phase (Psw) for a 3-level inverter can be expressed as:

$$Psw = Nc * (fsw / \pi) * Esw\_ref * (Ipk / Iref) * (Vdc\_sw / Vref)$$

Where Nc is the number of switches commutating per PWM transition (Nc = 2 for 3L-FC), Esw_ref is the datasheet reference switching energy at Iref and Vref, and Vdc_sw is the voltage blocked by the switch (Vdc/2 = 400V).

B. Device Calibration

The analytical model was calibrated using parameters from the selected Wolfspeed CAB450M12XM3 module. The reference switching energies are Eon = 25.4 mJ and Eoff = 7.51 mJ at Iref = 450A and Vref = 600V.

C. The Case for Adaptive Switching Frequency

The results reveal that operating at a fixed 20 kHz carrier frequency—while necessary for high-speed highway driving to maintain an acceptable frequency modulation ratio (mf = fsw / fe > 9)—is severely sub-optimal for urban driving. At 40 km/h, the high phase current (136 A) combined with a 20 kHz switching frequency generates 84.3 W of switching loss per phase.

Because switching loss scales linearly with fsw, reducing the carrier frequency to 5 kHz during urban driving cuts the switching loss to just 21.1 W per phase. This is permissible because the fundamental frequency at 40 km/h is only 184 Hz, yielding a safe modulation ratio (mf = 27) even at 5 kHz.

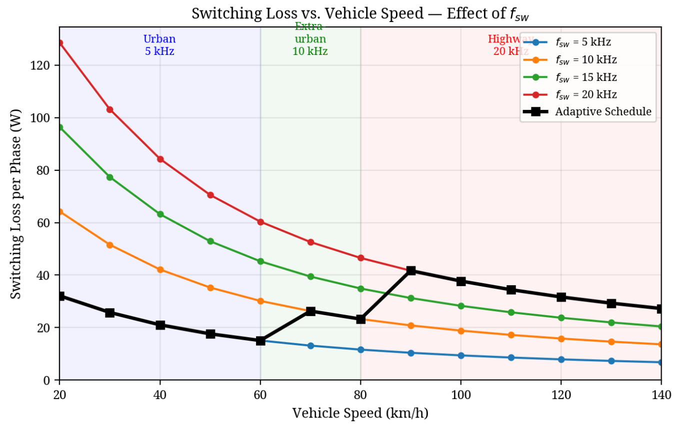

Fig. 2. Switching loss per phase across the vehicle speed range for various carrier frequencies. The black line indicates the adaptive schedule.

TABLE I. LOSS COMPARISON AT DESIGN POINTS

Metric	Urban (40 km/h)	Highway (120 km/h)
Electrical Freq (fe)	183.9 Hz	551.7 Hz
Peak Current (Ipk)	135.8 A	51.0 A
Fixed 20kHz Sw. Loss	84.3 W/phase	31.7 W/phase
Scheduled Sw. Loss	21.1 W/phase (5kHz)	31.7 W/phase (20kHz)

As detailed in Table I and Fig. 3, the adaptive schedule (5 kHz for <60 km/h, 10 kHz for 60-80 km/h, 20 kHz for >80 km/h) yields a massive 75% reduction in switching losses during urban driving, equating to a saving of nearly 190 W (3-phase) at 40 km/h.

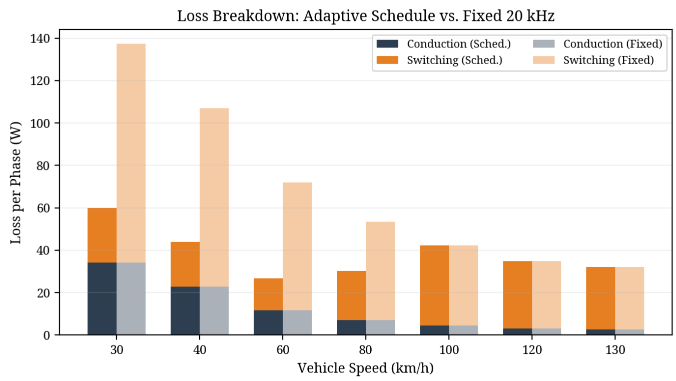

Fig. 3. Breakdown of conduction and switching losses comparing the adaptive schedule against a fixed 20 kHz baseline.

VI. TIME-DOMAIN SIMULATION RESULTS

To validate the operational stability of the design, a full time-domain simulation was constructed in Python using the datasheet-calibrated parameters.

A. Flying Capacitor Balancing and Pre-Charge

A critical operational requirement is the pre-charging of the 680 µF flying capacitor to prevent severe overvoltage across the inner switches at startup. The simulation incorporates a 20ms pre-charge sequence, linearly ramping Vfc to 400V before active PWM switching commences. As shown in Fig. 4, the PD-PWM balancing algorithm maintained the flying capacitor voltage exceptionally well, with a mean error of less than 0.04V from the 400V reference and a simulated urban ripple of 17.97V (matching the analytical 20V budget).

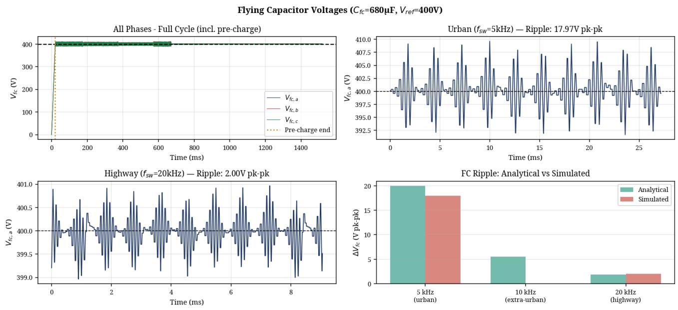

Fig. 4. Flying Capacitor voltage balancing demonstrating the 20ms pre-charge transient and steady-state ripple at urban and highway speeds.

B. Electrical Performance and THD

The inverter successfully synthesized the three-level output voltage. The Total Harmonic Distortion (THD) of the phase current, calculated using a Hanning-windowed FFT, was excellent: 1.92% for the urban segment (5 kHz) and 1.13% for the highway segment (20 kHz), as shown in Fig. 5.

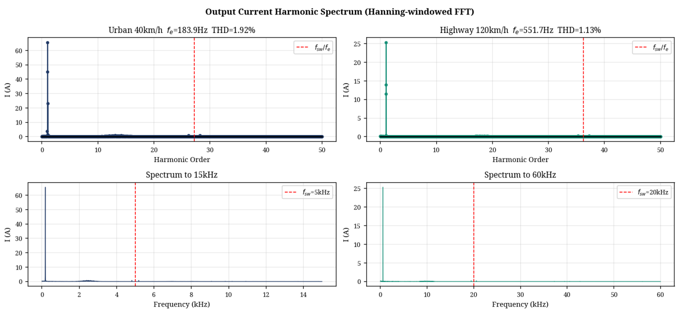

Fig. 5. Phase current harmonic spectrum and THD for urban (5 kHz) and highway (20 kHz) driving segments.

C. Thermal Analysis

The thermal performance of the CAB450M12XM3 module was evaluated using a 4-layer Foster network derived directly from the datasheet transient thermal impedance curves. With the adaptive switching frequency schedule, the maximum junction temperature reached only 46.4°C (assuming a 40°C ambient/coolant base), providing a massive 128.6°C margin to the module's 175°C limit. This confirms that the selected module and adaptive frequency strategy provide exceptional thermal reliability.

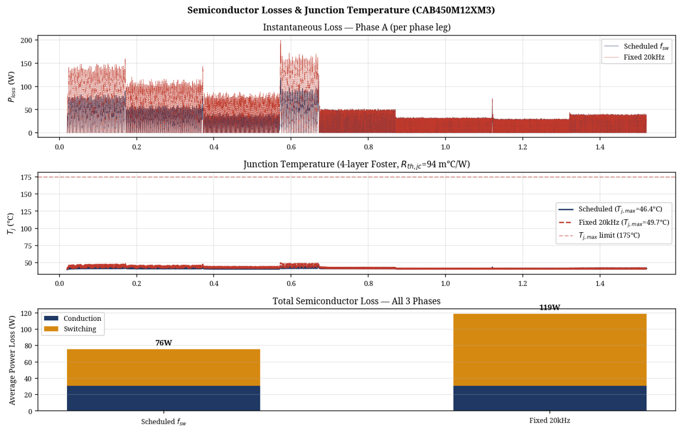

Fig. 6. Semiconductor junction temperature profile utilizing the 4-layer Foster thermal network model.

VII. CONCLUSION

This paper demonstrated that traction inverter design and optimization cannot be conducted in isolation from the vehicle architecture. The 3L-FC topology was selected over the NPC alternative due to its superior thermal symmetry and natural balancing capabilities under dynamic EV loads. Component selection was rigorously justified using industry standards for 800V architectures. By mapping mechanical vehicle speed to inverter electrical parameters, a comprehensive switching loss profile was generated using the Graovac-Purschel analytical method. Implementing a speed-adaptive switching frequency schedule reduced urban switching losses by 75%, saving 190 W (3-phase) while maintaining excellent FC voltage balancing, low THD (<2%), and safe thermal operation. This approach significantly improves overall drivetrain efficiency where EVs spend the majority of their operational life.

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